

Karan Singh Sodhi – Software Engineer

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Summary

Entry-level Software Engineer with hands-on experience building **full-stack and real-time web applications** using MERN and Next.js. Strong foundation in **data structures, RESTful API design, object-oriented programming, and backend performance optimization**. Experienced with **Docker**, and real-time systems using Socket.IO through internships and production-grade projects. Comfortable working in agile environments, debugging complex issues, and delivering reliable software.

Experience

Web Developer Intern, V'NSPIRE – Agra, India

- Shipped a **production-ready MERN application** with authentication, posting, and feed workflows, handling **50+ user-generated posts**.
- Delivered **real-time messaging** using Socket.IO, supporting **10+ concurrent users** with low-latency updates.
- Modeled and optimized **MongoDB collections** for users, posts, and messages, scaling to **1,000+ records** without performance degradation.
- Designed and exposed **10+ REST APIs** in Node.js and Express with clear contracts, validation, and error handling.
- Secured backend services using **JWT-based auth** and role checks, preventing unauthorized access to protected endpoints.
- Improved UI responsiveness and data flow through efficient client-side state management and render optimization.
- Worked closely with designers and peers in short iteration cycles, resolving bugs and shipping features on schedule.

Projects

Real-Time Collaboration Platform (Discord Clone)

Next.js 15, Socket.IO, LiveKit, PostgreSQL

Live | Code

- Engineered a scalable real-time communication system supporting low-latency messaging and voice/video streaming using WebSockets and SFU-based architecture.
- Designed normalized relational schemas with indexing, cascading constraints, and soft-deletion strategies to ensure data integrity and efficient querying.
- Implemented role-based access control, secure authentication flows, and server-side validation for isolated multi-server collaboration.
- Optimized event handling and state synchronization to minimize re-renders and reduce client-side performance overhead.

Scalable E-Commerce Platform

MongoDB, Express.js, React.js, Node.js, Stripe, Cloudinary

Live | Code

- Architected and deployed a production-ready full-stack eCommerce platform with modular REST APIs and scalable MERN architecture.
- Designed secure JWT-based authentication and role-based authorization (Admin/User), enforcing protected routes and access isolation.
- Integrated Stripe payment workflows with end-to-end order validation and Cloudinary-backed media optimization for efficient asset delivery.
- Optimized database indexing and query patterns in MongoDB, reducing response latency and improving overall application performance.

Education

Dr. A.P.J. Abdul Kalam Technical University

Bachelor of Technology in Computer Science and Engineering

Skills

Languages: C, C++, Java, Python, JavaScript, TypeScript

Frontend: React.js, Next.js, HTML, CSS

Backend: Node.js, Express.js, REST APIs, Asynchronous Processing, Authentication

Databases: PostgreSQL, MongoDB, Indexing, Query Optimization

Tools: Git, GitHub, Debugging, Problem Solving

CS Fundamentals: OOP, Data Structures, Algorithms, Operating Systems, Concurrency, DevOps

Certifications

- Full Stack Web Development (MERN)** — Sheriyans Coding School
- JavaScript (ES6+) & Data Structures** — Coursera
- AWS Cloud Practitioner & Fundamentals** — freeCodeCamp